

A Few Minor Differences Between Python 2 and Python 3



It's almost 11 years since Python 3 (Py3K) was released in December 2008.

The earliest version of Python was released in 1991 and since then, a few fundamental flaws had developed in the language. Python 3 was developed with the guiding principle of "Reducing feature duplication by removing old ways of doing things." Sadly, Python 3 is not backward compatible.

I had recently written a series of articles on Python for a children's science magazine. Though my programming experience is limited to Python 2, I decided to stick to Python 3 because the target audience is young. Python 3 is the future of this language. I came across a number of subtle differences between Python 2 and Python 3 while writing that series of articles. Well, this was to be expected as Python 3 is not backward compatible with Python 2. But still, I believe it will be useful to identify and list out such minor differences between Python 2 and Python 3 so that a programmer can easily migrate from the former to the latter.

Rather than dealing with the philosophical aspects of migrating from Python 2 to Python 3, let's consider practical situations in which a Python 2 programmer should exercise caution while programming in Python 3.

Differences with *print*

In Python 2, *print* is a statement whereas in Python 3, it is a function. Parentheses are not necessary in the

Python 2 *print* statement whereas in Python 3, it is necessary to do so. For example, the Python 2 *print* statement *print "Hello World"* should be written as *print ("Hello World")* in Python 3. Another difference with *print* in both versions occurs when a *print* statement should not move the cursor to a new line. Consider the small Python script *py1.py* shown below. Please note the comma after the first *print* statement.

```
print("Open Source "),
print("For You")
```

Figure 1 shows the execution of the script in Python 2 and Python 3, using the commands *python2* and *python3*, respectively. Python 2 prints the two

```
# python2 py1.py
Open Source For You
# python3 py1.py
Open Source
For You
# python py1.py
Open Source For You
#
```

Figure 1: Differences between *print* in Python 2 and Python 3

strings in a single line whereas Python 3 prints the two strings in two lines. From the figure we can also see that the command *python* in Linux still translates to the command *python2* which interprets the script as Python 2.

In order to print a string without moving the cursor to a new line in Python 3, we can use the *print* function as shown in the Python script *py2.py* given below.

```
print("Open Source ", end="")
print("For You")
```

Figure 2 shows that in Python 3, the two strings are printed in a single line, whereas in Python 2 the script *py2.py* results in a syntax error. This is yet another difference while using *print* in Python 2 and Python 3.

```
# python3 py2.py
Open Source For You
# python2 py2.py
File "py2.py", line 1
    print("Open Source ", end="")
                        ^
SyntaxError: invalid syntax
#
```

Figure 2: No new line *print* in Python 3